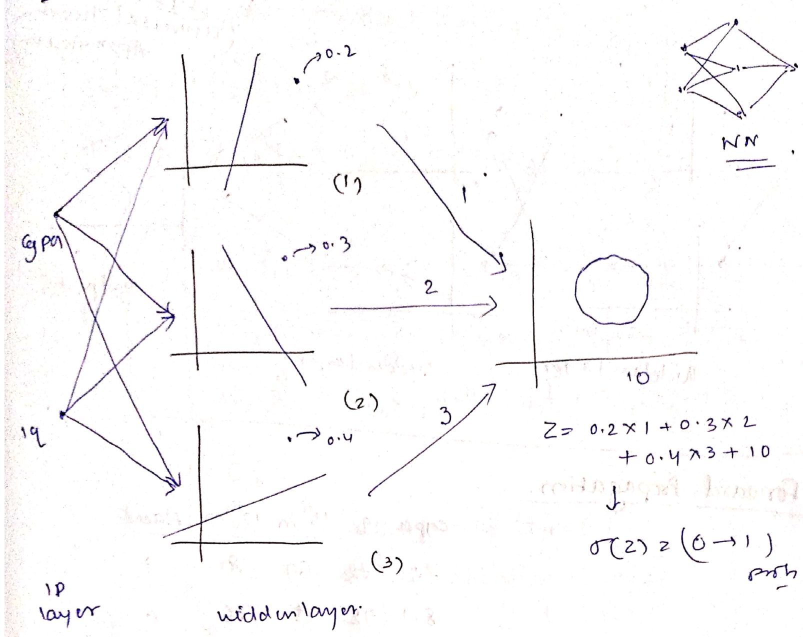
Adding nodes in hidden

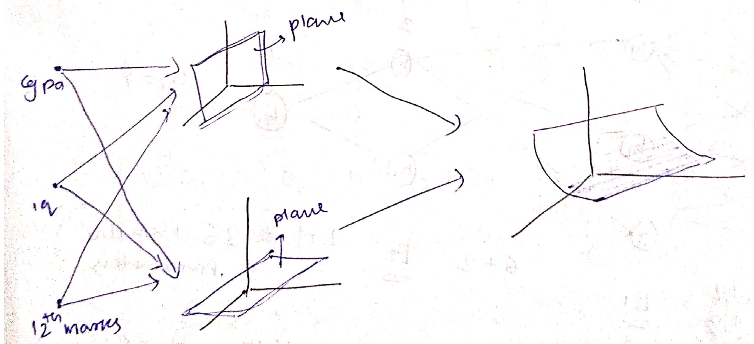


$Z = 0.2\times 1 + 0.3\times 2 + 0.4\times 3 + 10$

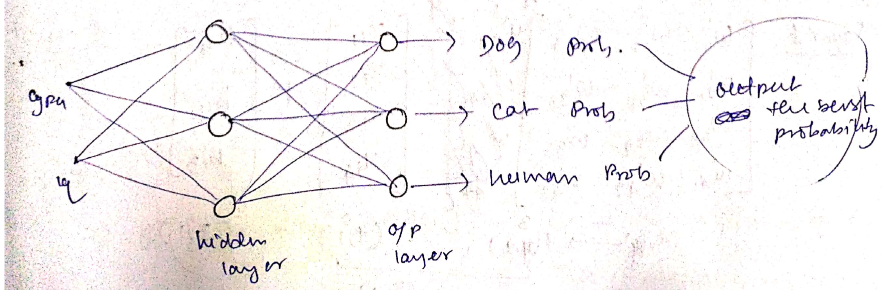
$\downarrow$

$\sigma(z) = (0 \rightarrow 1)$ prob.

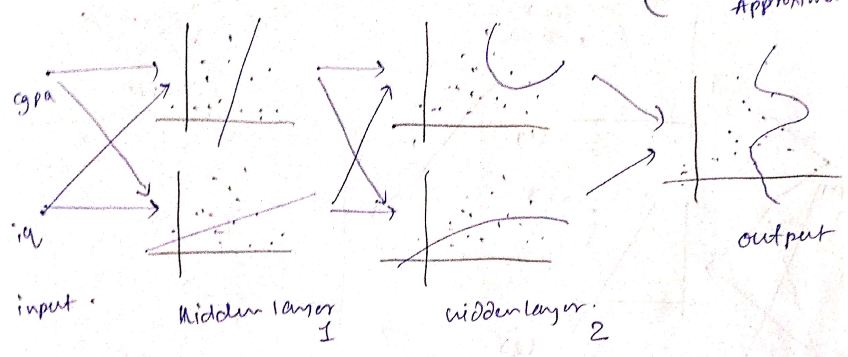
Adding Nodes in Input.



Adding nodes in output ~~layer~~ node.



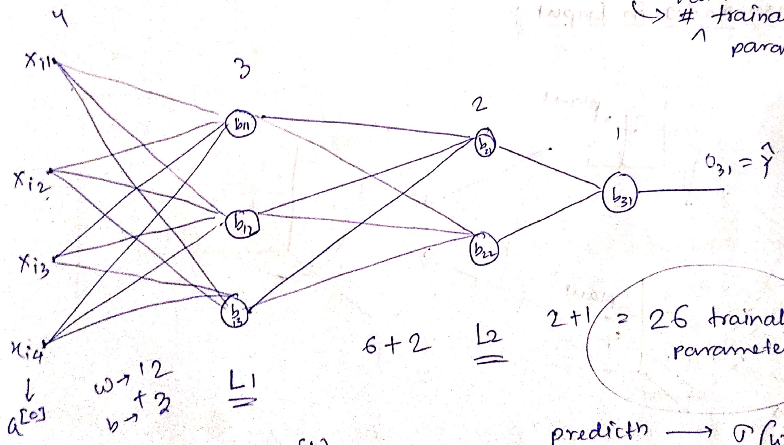
Deep Neural Network (Adding layers) → UFA (Universal function Approximator)



Forward Propagation.

cgpa	iq	10th m	12th m	placed
7.2	72	69	81	1
8.1	92	75	76	0

↳ # no. of trainable parameters



$w \rightarrow 12$, $b \rightarrow +3$ L1

$6+2$ L2

$2+1$ = 26 trainable parameters

predict → $\sigma(w^T x + b)$

layer 1

$$\underbrace{\begin{bmatrix} w^1_{11} & w^1_{12} & w^1_{13} \\ w^1_{21} & w^1_{22} & w^1_{23} \\ w^1_{31} & w^1_{32} & w^1_{33} \\ w^1_{41} & w^1_{42} & w^1_{43} \end{bmatrix}^T}_{w^{[1]}} \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$(4,3) \xrightarrow{T} (3,4)$ $(4,1)$ $(3,1)$

$(3,1)$

$$= \begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$\Rightarrow \sigma\left(\begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} + b_{11} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} + b_{12} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} + b_{13} \end{bmatrix} \right)$$

$$= \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} \rightarrow \text{output of layer 1.}$$

$= a^{[1]} \rightarrow$ activation of layer 1

layer 2.

$$\underbrace{\begin{bmatrix} w^2_{11} & w^2_{12} \\ w^2_{21} & w^2_{22} \\ w^2_{31} & w^2_{32} \end{bmatrix}}_{w^{[2]}}{}^{T} \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$(3,2) \xrightarrow{T} (2,3) \quad (3,1) \quad (2,1)$

$(2,1)$

$$= \begin{bmatrix} w^2_{11}O_{11} + w^2_{21}O_{12} + w^2_{31}O_{13} + b_{21} \\ w^2_{12}O_{11} + w^2_{22}O_{12} + w^2_{32}O_{13} + b_{22} \end{bmatrix} \cancel{+ \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}}$$

$$\Rightarrow \sigma\left(\begin{bmatrix} w^2_{11}O_{11} + w^2_{21}O_{12} + w^2_{31}O_{13} + b_{21} \\ w^2_{12}O_{11} + w^2_{22}O_{12} + w^2_{32}O_{13} + b_{22} \end{bmatrix} \right) = \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} \rightarrow \text{Output of layer 2}$$

$= a^{[2]} \rightarrow$ activation of layer 2

layer 3

$$\underbrace{\begin{bmatrix} w^3_{11} \\ w^3_{21} \end{bmatrix}}_{w^{[3]}}{}^{T} \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$(2,1) \xrightarrow{T} (1,2) \quad (2,1) \quad (1,1)$

$$\Rightarrow \sigma\left(\begin{bmatrix} w^3_{11}O_{21} + w^2_{21}O_{22} + b_{31} \end{bmatrix} \right) = O_{31} = \hat{Y}_i \rightarrow \text{final output}$$

$= a^{[3]} \rightarrow$ activation 3.

$a^{[1]} = \sigma\left(a^{[0]}w^{[1]} + b^{[1]}\right)$

$a^{[2]} = \sigma\left(a^{[1]}w^{[2]} + b^{[2]}\right)$

$a^{[3]} = \sigma\left(a^{[2]}w^{[3]} + b^{[3]}\right) \nearrow$ final o/p

$$\sigma\left(\sigma\left(\underbrace{\sigma\left(a^{[0]}w^{[1]} + b^{[1]}\right)}_{a^{[1]}} w^{[2]} + b^{[2]}\right) w^{[3]} + b^{[3]}\right)$$

$a^{[2]}$

$a^{[3]}$

18/07/25.

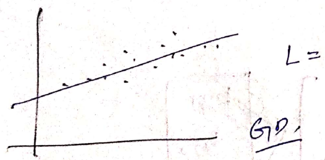
What is Loss fn?

Loss fn is a method of evaluating how well your algorithm is modelling your dataset.

high → poor

small → great

L (parameters) ← $f(x) = x^2 + 2$



$L = (Y_i - \hat{Y}_i)^2$

$\hat{Y}_i = mx_i + b$

$L(m, b) = \min.$

GD.

Why is Loss fn important?

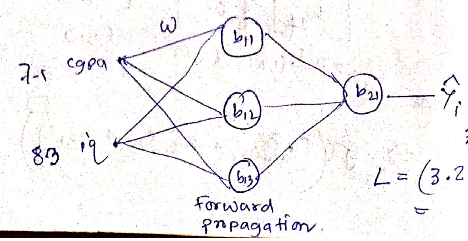
[You can't improve what you can't measure]

Peter Drucker

Loss function ⟶ eye of ML algorithm.

Loss fn in Deep Learning?

cgpa	iq	package (LPA)
7.1	83	3.2
8.5	91	4.5
6.3	102	6.1
5.1	89	2.7



3.7

$L = (3.2 - 3.7$